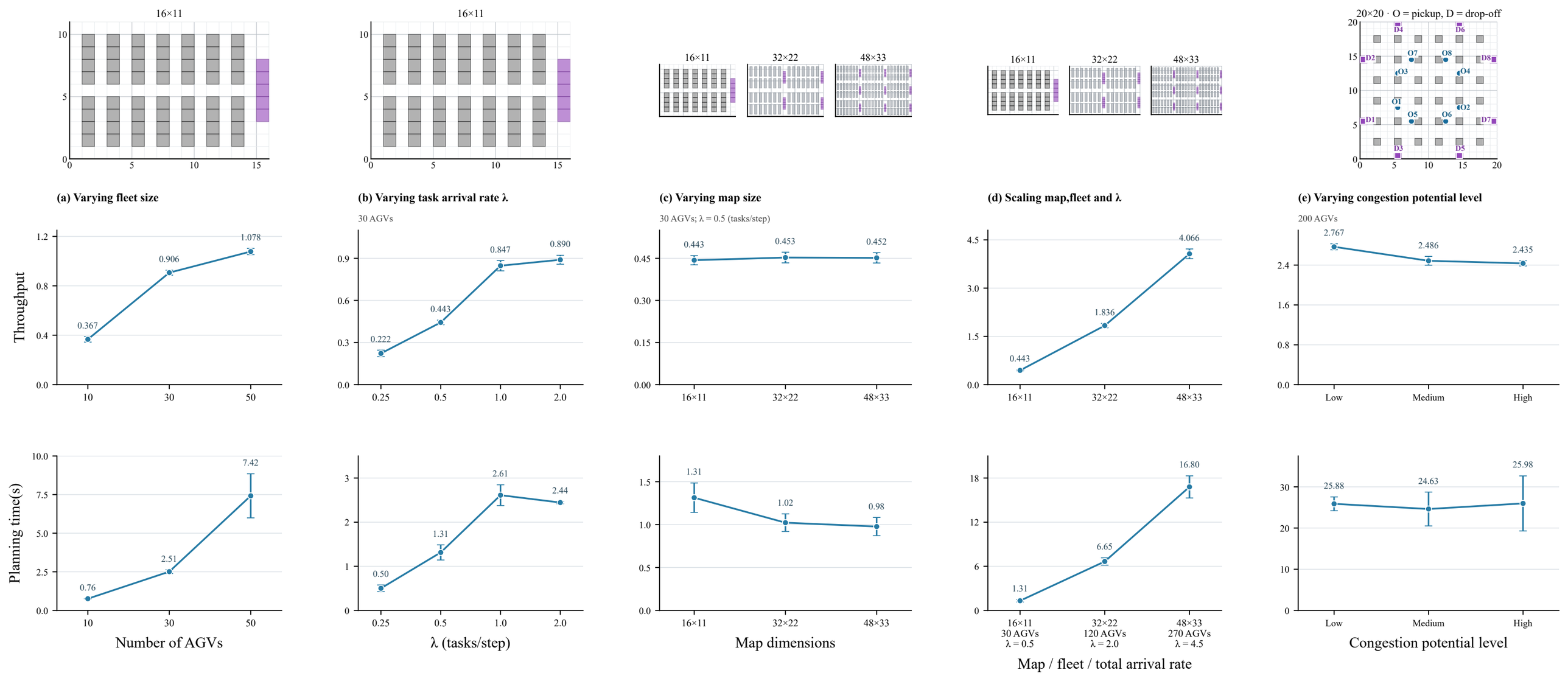

16×11
16×11
16×11
32×22
48×33
16×11
32×22
48×33
20×20 · O = pickup, D = drop-off
(a) Varying fleet size
(b) Varying task arrival rate λ
(c) Varying map size
(d) Scaling map,fleet and λ
(e) Varying congestion potential level
30 AGVs
30 AGVs; λ = 0.5 (tasks/step)
200 AGVs
Throughput
Planning time(s)
0.367
0.906
1.078
0.222
0.443
0.847
0.890
0.443
0.453
0.452
0.443
1.836
4.066
2.767
2.486
2.435
0.76
2.51
7.42
0.50
1.31
2.61
2.44
1.31
1.02
0.98
1.31
6.65
16.80
25.88
24.63
25.98
Number of AGVs
λ (tasks/step)
Map dimensions
Map / fleet / total arrival rate
Congestion potential level
Low
Medium
High
16×11
30 AGVs
λ = 0.5
32×22
120 AGVs
λ = 2.0
48×33
270 AGVs
λ = 4.5